MATHEMATICS PROJECT WORK RECORD :: FA-3

Topic: Coordinate Geometry

1. PRELIMINARY INFORMATION

Name of the Student:	.....
Class & Section:	.....
Roll Number:	.....
Subject:	**Mathematics**
Topic:	**Coordinate Geometry**

2. AIM

To verify the distance formula on a coordinate plane and calculate the area of a right-angled triangle formed by coordinate points using both the basic formula and the coordinate geometry formula to demonstrate that both yield identical results.

3. OBJECTIVES

- Plot vertices A(1, 1), B(5, 1), and C(5, 4) accurately on a Cartesian coordinate plane.
- Calculate lengths AB, BC, and AC using the distance formula: $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$.
- Compute triangle area using Method 1 (Basic): Area = (1/2) × base × height.
- Compute triangle area using Method 2 (Coordinate): Area = $(1/2)\,|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$.
- Verify that both analytical and geometric formulas evaluate to exactly 6 sq. units.

4. MATERIALS REQUIRED

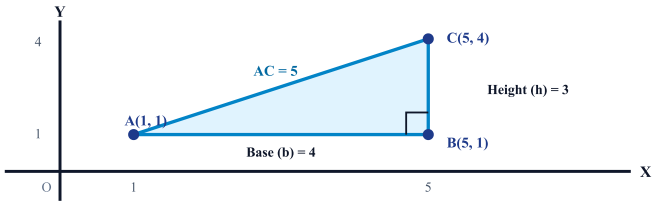
Graph sheet, 30 cm ruler, pencil, eraser, pen, and Class 10 Mathematics textbook.

5. PROCEDURE

1. Drawn orthogonal axes X and Y intersecting at origin O(0, 0).
2. Plotted vertices A(1, 1), B(5, 1), and C(5, 4) to construct right-angled ΔABC.
3. Determined base length AB and perpendicular height BC using the distance formula.
4. Evaluated area via basic formula: (1/2) × base × height.
5. Substituted vertex coordinates into the coordinate area formula and simplified.
6. Compared both outcomes to confirm consistency.

6. OBSERVATIONS AND TABULATION

Side	Endpoints	Distance Formula Working	Length
Base (AB)	A(1, 1), B(5, 1)	$\sqrt{[(5-1)^2 + (1-1)^2]} = \sqrt{(16+0)}$	4 units
Height (BC)	B(5, 1), C(5, 4)	$\sqrt{[(5-5)^2 + (4-1)^2]} = \sqrt{(0+9)}$	3 units
Hypotenuse (AC)	**A(1, 1), C(5, 4)**	$\mathbf{\sqrt{[(5-1)^2 + (4-1)^2]} = \sqrt{(16+9)}}$	**5 units**



Method 1: Basic Formula

Area = (1/2) × base × height
= (1/2) × AB × BC
= (1/2) × 4 × 3
= 12 / 2 = **6 sq. units**

Method 2: Coordinate Formula

Area = (1/2) $|x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)|$
= (1/2) $|1(1-4) + 5(4-1) + 5(1-1)|$
= (1/2) $|1(-3) + 5(3) + 5(0)|$
= (1/2) $|-3 + 15 + 0|$ = (1/2) $|12|$ = **6 sq. units**

Verification: Method 1 (Basic) = Method 2 (Coordinate) = 6 sq. units

7. CONCLUSION

- The distance formula verifies side lengths AB = 4, BC = 3, and hypotenuse AC = 5.
- Both the basic formula ((1/2) × b × h) and the coordinate geometry formula yield identical area (**6 sq. units**).

8. ACKNOWLEDGEMENT

I express my sincere thanks to my **Mathematics Teacher** for continuous guidance throughout this project.

9. RESOURCES

Telangana SCERT Class 10 Mathematics Textbook (Topic: *Coordinate Geometry*).

Signature of Student
Date: ____/____/2026

Signature of Mathematics Teacher
Verified